Tutorial - 6

10/3/26

Ridge Regression (linear model)

Study hours	Attendance %	previous score	sleep Hours	final grade
3				

Bedroom (x_1)	Age (years) (x_2)	price (y)
3	5	2
3	3	3

linear eqn

$2 = c + 3m_1 + 5m_2$

$3 = c + 3m_1 + 3m_2$

$$\underbrace{\begin{pmatrix} 2 \\ 3 \end{pmatrix}}_{y} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{x} \underbrace{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}_{w}$$

$$w = (x^T x + \lambda I)^{-1} (x^T y)$$

Take $\lambda = 1$

$$x = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} \cdot x^T = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}$$

$$x^T x = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{pmatrix}$$

$$x^T x = \begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$(X^T X + \lambda I)$

$$= \begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix} + 1 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{pmatrix}$$

$$(X^T X + \lambda I)^{-1} = \frac{\text{Adj}\,(X^T X + \lambda I)^{-1}}{\det (X^T X + \lambda I)^{-1}}$$

$$= \frac{1}{95} \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix}$$

$$X^T Y = \begin{pmatrix} 1 & 3 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ 19 \end{pmatrix}$$

$$w = (X^T X + \lambda I)^{-1} \cdot (X^T Y)$$

$$= \frac{1}{95} \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$= \frac{1}{95}\begin{bmatrix} 23 \\ 69 \\ -1 \end{bmatrix}$$

$$\approx \begin{array}{c} \begin{bmatrix} 0.242 \\ 0.726 \\ -0.0105 \end{bmatrix} \end{array} \begin{array}{l} c \\ m_1 \\ m_2 \end{array}$$

$$\boxed{\hat{y}_{pred} = c + m_1 x_1 + m_2 x_2}$$

$$= 0.242 + 0.726 x_1 - 0.0105 x_2$$

$$= 0.242 + 0.726(3) - 0.0105(5)$$

$$\hat{y}_1 = 2.3675 \quad , \quad y_1 = 2$$

$$\hat{y}_2 = 2.3885 \quad , \quad y_2 = 3$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$$= \frac{1}{2}\left((2 - 2.3675)^2 + (3 - 2.3885)^2\right)$$

$$= \frac{1}{2} \times 0.5099885$$

$$\approx 0.255$$

$$\bar{y} = \frac{5}{2} = 2.5$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

$$= 1 - \frac{0.255 \times 2}{(2-2.5)^2 + (3-2.5)^2}$$

$$R^2 = -0.042$$